

AGROSPEAR CROPCARE CO., LTD.

INSECTICIDE 4-STAGE QA/QC LIFECYCLE REPORTS & BATCH RELEASE DOSSIER

杀虫剂全生命周期品控（4阶段检验报告、四级关卡放行确认表与偏差CAPA处置报告）

Batch Record ID: BMR-INS-20260826-03
Target Formulation: Chlorantraniliprole 200g/L SC
Batch Quantity: 12,000 L (1L x 12/CTN)
Manufacturing Date: 2026-08-26
QA Authorization: Fully Audited & Released

STAGE 1: INCOMING QUALITY CONTROL (IQC) INSPECTION REPORT / 阶段一：原辅料与包材进厂检验报告

IQC Report No.: IQC-INS-20260820-01		Sampling Date: 2026-08-20	Inspector: Zhao Lei (赵磊)	Overall Verdict: PASSED / 合格入库
Material / Item	Supplier & Lot No.	Standard Specification	Actual Test Result	Verdict
Chlorantraniliprole TC 氯虫苯甲酰胺原药	Agrospear Tech Mat. Lot: TC-20260815-98	• AI Purity $\geq 97.50\%$ (HPLC) • Related Impurity IN-EQW78 $\leq 0.20\%$ • Moisture $\leq 0.50\%$, Acidity $\leq 0.30\%$	• AI Purity: 98.42% • IN-EQW78: 0.08% • Moisture: 0.18%, Acidity: 0.08%	Conforms
Polycarboxylate Dispersant 聚羧酸盐分散助剂	Specialty Surfactants Lot: ADJ-20260818-04	• Solid Content: $40.0 \pm 2.0\%$ • pH: 6.5 – 8.5 (1% aq.) • Surface Tension ≤ 32.0 mN/m	• Solid Content: 40.8% • pH: 7.2 • Surface Tension: 29.6 mN/m	Conforms
Fluorinated HDPE Bottles (1L) 1升多层氟化高阻隔瓶	Polymer Pack Co. Lot: PK-20260819-B1	• Wall Thickness ≥ 0.90 mm • Drop Impact (1.8m, -10°C): 0% break • Induction Foil Compatibility: 100%	• Min Wall: 0.96 mm • 1.8m Drop: 0/20 cracked • Foil Seal: Passed 70 kPa vacuum	Conforms

STAGE 2: IN-PROCESS QUALITY CONTROL (IPQC) INSPECTION REPORT / 阶段二：生产制程过程巡检报告

IPQC Record No.: IPQC-INS-20260826-02		Process Line: SC-Milling & Filling Line #2	IPQC Officer: Sun Qian (孙倩)	Process State: STABLE / 制程受控
Process Checkpoint	Critical Control Parameter (CCP)	In-Line Measurements (Continuous / 30m)	Evaluation & Action	
High-Shear Pre-mixing 高剪切初浆配制	• Temperature $< 40^{\circ}\text{C}$ • Uniform dispersion without agglomerates	• Slurry Temp: $31.5^{\circ}\text{C} - 34.2^{\circ}\text{C}$ • Visual: Uniform creamy suspension	Normal / 正常	
Horizontal Bead Milling 两段卧式砂磨细化	• Mill Chamber Outlet Temp $\leq 38^{\circ}\text{C}$ • Chamber Pressure < 0.25 MPa • Target Fineness: D50 $< 2.0\mu\text{m}$, D90 $< 4.5\mu\text{m}$	• Outlet Temp: 32.4°C (Peak: 34.8°C) • Pressure: 0.14 MPa • Laser Particle Size: D50 = $1.38\mu\text{m}$, D90 = $2.85\mu\text{m}$	In-Spec / 粒径达标	
Structuring & Thickening 胶体触变体系构建	• Viscosity: 400 – 1200 mPa·s (Brookfield #2/30rpm) • Thixotropic Ratio ≥ 2.5	• Viscosity: 760 mPa·s • Thixotropic Index: 3.1	Non-settling / 抗沉降	
Automated Filling & Foil Sealing 自动灌装与铝箔封口	• Net Volume: 1000 ± 5 mL (100% In-line Weigher) • Induction Seal: 100% hermetic (Vacuum -70 kPa) • 2D Laser Batch Coding legibility	• Average Filling: 1002.4 mL (Min: 998, Max: 1004) • Torque: 2.8 N·m, Vacuum leak: 0/50 defect • 100% Scanner readable	Passed / 密封合格	

STAGE 3: FINAL QUALITY CONTROL (FQC) & OFFICIAL COA / 阶段三：成品出厂终检分析报告单 (COA)

Product: Chlorantraniliprole 200g/L SC		Batch: AS-INS-20260826-03	Mfg: 2026-08-26	Exp: 2028-08-25
Test Item / 检测项目	Specification Limit / 规格标准	Test Result / 实测结果	Method / 方法	Verdict
Appearance / 外观性状	Off-white homogeneous flowable suspension	Conforms (类白色均质流体)	Visual	Pass
Active Content / 含量	200.0 ± 10.0 g/L (190.0 – 210.0 g/L)	201.8 g/L	HPLC-DAD	Pass
pH Value (1% aq. solution)	5.5 – 8.5	6.72	CIPAC MT 75.3	Pass
Suspensibility / 悬浮率	$\geq 90.0\%$ (CIPAC Standard Water D)	96.4 %	CIPAC MT 184	Pass
Particle Size / 粒度 (D90)	$\leq 4.5 \mu\text{m}$	$2.85 \mu\text{m}$ (D50: $1.38 \mu\text{m}$)	Laser Sizer	Pass
Wet Sieve / 湿筛 (75 μm)	$\geq 99.0\%$ through	99.92 %	CIPAC MT 185	Pass
Persistent Foam / 起泡性	≤ 40 mL after 1 min	12 mL	CIPAC MT 47.2	Pass
Pourability / 倾倒性	Residue $\leq 4.0\%$, Rinsed residue $\leq 0.5\%$	Residue: 1.8 %, Rinsed: 0.18 %	CIPAC MT 148.1	Pass
Accelerated Heat Stability	$54 \pm 2^{\circ}\text{C}$ for 14d, AI degradation $\leq 5.0\%$	Degradation: 0.85%, Susp: 95.2%	CIPAC MT 46.3	Pass
Cold Stability / 冷储	$0 \pm 2^{\circ}\text{C}$ for 7d, no separation / crystals	Conforms / 无析出	CIPAC MT 39.3	Pass

STAGE 4: SAMPLE RETENTION & TRACEABILITY ARCHIVE RECORD / 阶段四：留样归档与全流程数字化溯源记录

Retention Checkpoint	Operational Protocols & Archival Data	Compliance Status
Retained Sample Vault 恒温恒湿留样室归档	• Archival Quantity: 500 mL x 6 bottles (Commercial package) • Vault Location: Shelf A-08, Vault Room #2 ($20 \pm 2^{\circ}\text{C}$, RH $50 \pm 5\%$) • Retention Period: Expiry Date + 1 Year (Archived until: 2029-08-25)	Archived & Tagged / 完好归档
2D QR Serialization 一物一码数字化溯源	• Single bottle serialized range: AGR202608260300001 – AGR202608260312000 • Cloud ERP linking: TC Lot (TC-20260815-98) → Reactor #2 → Filling Machine #2 → Pallet Barcode → Customs Export Invoice	100% Traceable / 双向可溯

FOUR-LEVEL QUALITY RELEASE AUTHORIZATION FORM / 杀虫剂四级关卡放行确认表

Release Gate / 关卡	Mandatory Verification Criteria / 核心放行审核条件	Status / 状态	Sign & Date / 审核签章
Gate 1: Raw Material IQC 一级关卡：原料放行	Chlorantraniliprole TC purity ≥ 97.5%, adjuvant solid content & packaging pressure tests 100% passed.	Verified & Released	IQC Lead: Zhao Lei 2026-08-20
Gate 2: Process IPQC 二级关卡：制程放行	Milling chamber temp < 38°C, particle size D90 < 4.5µm, volume accuracy & foil vacuum seal passed.	Verified & Released	IPQC Lead: Sun Qian 2026-08-26
Gate 3: Lab FQC (COA) 三级关卡：实验室终检	HPLC active content 201.8 g/L (in-spec), suspensibility 96.4%, 14d accelerated heat & cold storage passed.	Verified & Released	Lab Supervisor: Li Ming 2026-08-27
Gate 4: Packaging & QR 四级关卡：追溯放行	Physical sample retention archived, 2D QR serialization cloud uploaded, customs shipping dossier verified.	Verified & Released	QA Officer: Zhang Wei 2026-08-27

FINAL QA DISPOSITION / 质量部最终放行决议：
Batch No. AS-INS-20260826-03 has passed all 4-Level Quality Gates with zero critical deviations. **APPROVED FOR COMMERCIAL RELEASE & EXPORT.**
该批次杀虫剂已全项通过四级质量关卡，无关键质量缺陷，准予出厂放行并签发对外商检报关文件。

DEVIATION HANDLING & CAPA INVESTIGATION PROTOCOL / 生产偏差处置与纠正预防措施 (CAPA) 规程

OUT-OF-SPECIFICATION (OOS) PROTOCOL: Any parameter excursion triggers automated ERP physical quarantine (Red Tag). Production is halted immediately. A cross-functional team (QA, Production, Formulation R&D) initiates root cause investigation.
超标 (OOS) 处置规程：生产或检测指标若发生偏离，系统立即对该批产品自动加锁并张贴物理红卡隔离，相关工段即刻停产。由质量、生产、研发成立联合调查组开展根本原因排查。

Failure Mode / 典型偏差	Root Cause Investigation / 根因调查方向	Corrective & Preventive Action (CAPA) / 纠正与预防措施	Authorization
Active Assay Deviation 有效成分含量偏离 (±5%)	- Raw TC assay fluctuation - Formulation weigh scale drift - Incomplete dissolution/dispersion	- Re-calibrate micro-dosing load cells with F1 standard weights. - Secondary titration & HPLC double validation. - Formulate precise compensating batch rework protocol.	QA Director 质量总监
SC Coarse Particle (D90 > 5.0µm) 悬浮剂研磨粒度超标	- Zirconia bead wear/loss - High flow-rate during milling - Cooling chiller temp spike	- Replace & top-up 0.8mm Yttrium zirconia beads. - Throttle pump rate & reduce jacket temp to < 15°C. - Re-mill batch for additional 2 cycles.	Production Head 车间主任
Induction Foil Seal Leak 铝箔封口微渗漏	- High-frequency coil power drift - Bottle neck uneven flash - Capper torque insufficient	- Adjust induction power generator & conveyor speed. 100% vacuum chamber re-inspection (-70 kPa). - Reject defective packaging lots back to vendor.	Packaging QA 包材主管

Prepared by (QA Engineer):
Zhang Wei (张伟)
Date: 2026-08-27

Audited by (Plant Director):
Li Jianguo (李建国)
Date: 2026-08-27

Approved by (QA Director):
Dr. Alex Chen (陈博士) [Approved]
Date: 2026-08-27